Contents available at ScienceDirect

Diabetes Research and Clinical Practice

journal homepage: www.elsevier.com/locate/diabres

A systematic review of interventions targeting primary care or community based professionals on cardio-metabolic risk factor control in people with diabetes

S. Seidu [a,b,*], N.S. Walker [c], D.H. Bodicoat [a,b], M.J. Davies [a,b], K. Khunti [a,b]

[a] Leicester Diabetes Centre, Leicester General Hospital, Gwendolen Road, Leicester LE5 4WP, UK
[b] Diabetes Research Centre, University of Leicester, Leicester General Hospital, Gwendolen Road, Leicester LE5 4WP, UK
[c] Department of Health Sciences, University of Leicester, 22-28 Princess Road West, Leicester LE1 6TP, UK

ARTICLE INFO

Article history:
Received 29 July 2015
Accepted 14 January 2016
Available online 21 January 2016

Keywords:
Diabetes
Interventions
Primary care
Health care professional

ABSTRACT

Objective: To review the interventions targeting primary care or community based professionals on glycaemic and cardiovascular risk factor control in people with diabetes.

Research design and methods: A systematic review of randomised controlled trials evaluating the effectiveness of interventions targeting primary care or community based professionals on diabetes and cardiovascular risk factor control. We conducted searches in MEDLINE database from inception up to 27th September 2015. We also retrieved articles related to diabetes from the Cochrane EPOC database and EMBASE and scanned bibliographies for key articles.

Results: There was heterogeneity in terms of interventions and participants amongst the 30 studies (39,439 patients) that met the inclusion criteria. Nine of the studies focused on general or family practitioners, five on pharmacists, three on nurses and one each on dieticians and community workers. Twelve studies targeted multi-disciplinary teams.

Educational interventions did not seem to have a positive impact on HbA1c, systolic blood pressure or lipid profiles. The use of telemedicine, clinician reminders and feedback showed mixed results but there was a level of consistency in improvement in HbA1c when multifaceted interventions on multidisciplinary teams were implemented. Targeting general or family physicians was largely ineffective in improving the cardiovascular risk factors considered, except when using a computer application on insulin handling of type 2 diabetes or customised simulated cases with feedbacks. Similarly, interventions targeting nurses did not improve outcomes compared to standard care.

Conclusions: Multifaceted professional interventions were more effective than single interventions targeting single primary or community care professionals in improving glycaemic control.

© 2016 Elsevier Ireland Ltd. All rights reserved.

* Corresponding author at: University of Leicester, Diabetes Research Centre, Leicester General Hospital, Leicester Diabetes Centre, Gwendolen Road, Leicester LE5 4WP, UK. Tel.: +44 07525191097; fax: +44 07525191097.
E-mail address: sis11@le.ac.uk (S. Seidu).

http://dx.doi.org/10.1016/j.diabres.2016.01.022
0168-8227/© 2016 Elsevier Ireland Ltd. All rights reserved.

1. Background

Diabetes has now been recognised as an epidemic globally. The prevalence of diabetes is estimated to be 382 million worldwide and is expected to rise to 592 million by 2035 [1]. The burden of the disease and its complications are outstripping health care systems globally [2]. The rising prevalence coupled with increasing life expectancy has prompted the transfer of some diabetes patients from specialist care to primary care [3]. Community or primary care diabetes management is therefore a logical focal point for implementing strategies that improve the care of people with diabetes. The pay-for-performance initiative started in the UK in 2004 is probably the biggest quality improvement (QI) strategy [4] and has yielded some obvious improvements in the care of people with diabetes. However these benefits have reached a plateau across the population [5] and additionally it has not succeeded in reducing the variations in care [6,7]. The disconnect between the evidence generated by researchers and the use of this best available evidence by health care providers in the field of diabetes is due to the complex interplay of factors ranging from poor co-ordination between primary and secondary care and also the difficulty of effecting lifestyle change [8].

Despite the ready availability of evidence-based diabetes guidelines and continuing professional development programmes, the gap between evidence-based best practice and quality of care is widely recognised [9]. Effective strategies [10–13] that promote the adoption of the best practice clinical guidelines for primary care management of diabetes therefore need examination. Continuing medical education is a commonly employed mechanism to improve primary care physicians' clinical practice [12].

Even though several publications and reviews have been published on interventions targeted at primary care professionals to improve outcomes, not many have been published in the field of diabetes. The only systematic review looking at just primary care professionals in the community settings was published in 2001 [14]. It is therefore important to update and critically evaluate the evidence for these interventions in view of the global burden of the disease. If they are found to be ineffective, payers will need to explore the reasons and perhaps invest the limited resources elsewhere. On the other hand, if some interventions are found to be effective, best practice could be shared for implementation in various countries to equip primary care and community based staff with the skill and capacity to deal with this ''left shift''. Our aim is to conduct a systematic review with emphasis on randomised controlled trials and considering risk factor control including glycaeted haemoglobin (HbA1c), systolic blood pressure (SBP) and total and LDL-cholesterol.

2. Methods

2.1. *Search strategies*

We combined text words and quality improvement Medical Subject Headings related to diabetes and our stated outcomes to search the MEDLINE database from inception up to 27th September 2015. We also retrieved all articles related to diabetes care from the Cochrane EPOC database and EMBASE, and scanned bibliographies from key articles. The detailed search terms are shown in Appendix 1. The review was registered on PROSPERO (registration number 2014013448) and is within the scope of the Centre for Reviews and Dissemination.

2.2. *Study selection*

S.S screened the titles and abstracts of all retrieved articles. Two reviewers (S.S. and N.S.W.) screened the full text of papers selected. We included any study on patients with type 1 or type 2 diabetes in primary care or community settings in which the intervention was either Clinician Education, Clinician Reminders, Team Changes, Case Management, Electronic Patient Registry, telemedicine, Audit and Feedback. We excluded interventions targeting only patients (for example, structured patient education, self-management or patient reminders) as there is now good evidence on the effectiveness of these interventions on various health outcomes [15–19].

We however included these if the interventions involved at least one component directed at clinician behaviour or organisational change. Multiple interventions or interventions that involved using the multidisciplinary team members, expansion of their roles or addition of other professionals such as pharmacists and nutritionists to the primary health care teams, were classified as multi-component interventions. Only studies looking at the cardiovascular risk factors HbA1c, SBP and total or LDL-cholesterol were included. Disagreements were resolved by consensus.

2.3. *Data extraction*

Data extraction was conducted using the Cochrane Group Data collection form for intervention reviews: RCTs only (version 3, April 2014), modified to meet the selection criteria by reducing the risk of bias. The information on the data collection forms included: author, year of publication, participants, location/setting, study design, interventions, outcomes reported and description of any other important sources of bias. The effects of the various interventions on the selected outcomes were considered in relation to the various health care professionals. Study results on HbA1c, cholesterol and blood pressure were classified as 'positive' if they were statistically significant or if the authors reported it so in the absence of inferential statistics. Because of substantial heterogeneity due to large differences in clinical or methodological nature between the studies, we decided a priori not to pool the data from the studies in a meta-analysis. Instead, we based a qualitative assessment of the effects on the quality of the studies. Quality assessment was based on the Cochrane risk of bias tool. Important sources of bias were considered around the following domains: random sequence generation (selection bias), blinding of outcome assessment (detection bias), incomplete outcome data (attrition bias) and selective outcome reporting (reporting bias).

3. Results

A summary of the literature search results is shown in Fig. 1. The search of the computerised databases identified 2289 citations and another 10 citations after scanning bibliographies for key articles on the subject. After excluding duplicates and studies clearly not related to the objective of our review, 1760 articles were considered for screening and 157 articles were eligible for full-text review. Overall a total of 30 articles met the inclusion criteria for this review.

4. Study characteristics and participants

Tables 1 and 2 show a summary of all the studies included. The studies included 13 randomised controlled trials [20–31] and 17 cluster randomised controlled trials [32–48]. In five studies the interventions were carried out on the whole diabetes population; in the remaining 25 studies only people with type 2 diabetes were included. Ten studies were conducted in the USA [21,23,25,29,33,36,38,39,43,45], four in the UK [26,28,32,41], four in Australia, [20,34,46,48], three in Denmark [24,40,42], two in the Netherlands [22,49] and one each in Spain [35], Ireland [37], Belgium [50], American Samoa [22], Taiwan [31] and Israel [44]. All the studies were published between 2003 and 2014.

Across all the studies a total of 39,439 patients were randomised and followed up for a mean duration of 16.7 months, ranging from 6 to 72 months. Nine of the studies focused on general practitioners or family practitioners [33–37,39–41,50]. Of these, three studies considered some form of feedback as an intervention; either as electronic feedback [40], feedback benchmarked against other centres [50] and customised simulated patient with feedbacks [38]. A further four of the studies targeting general practitioners considered education [33,34,37,41] and the remaining 2 studies included telemedicine interventions [35,36].

Three studies focused on interventions aimed at practice nurses; one of which was face to face review of glucose levels, blood pressure and lipid profile according to a specified protocol [49] another used telephone reminders [46] and in the third, practice nurses were offered a self-determination theory–based course including communication training [24]. Five studies included interventions aimed at pharmacists [20,23,25,28,48] who provided enhanced care to patients such counselling, advising, diabetes education, using applied algorithms for managing glucose control and decreasing

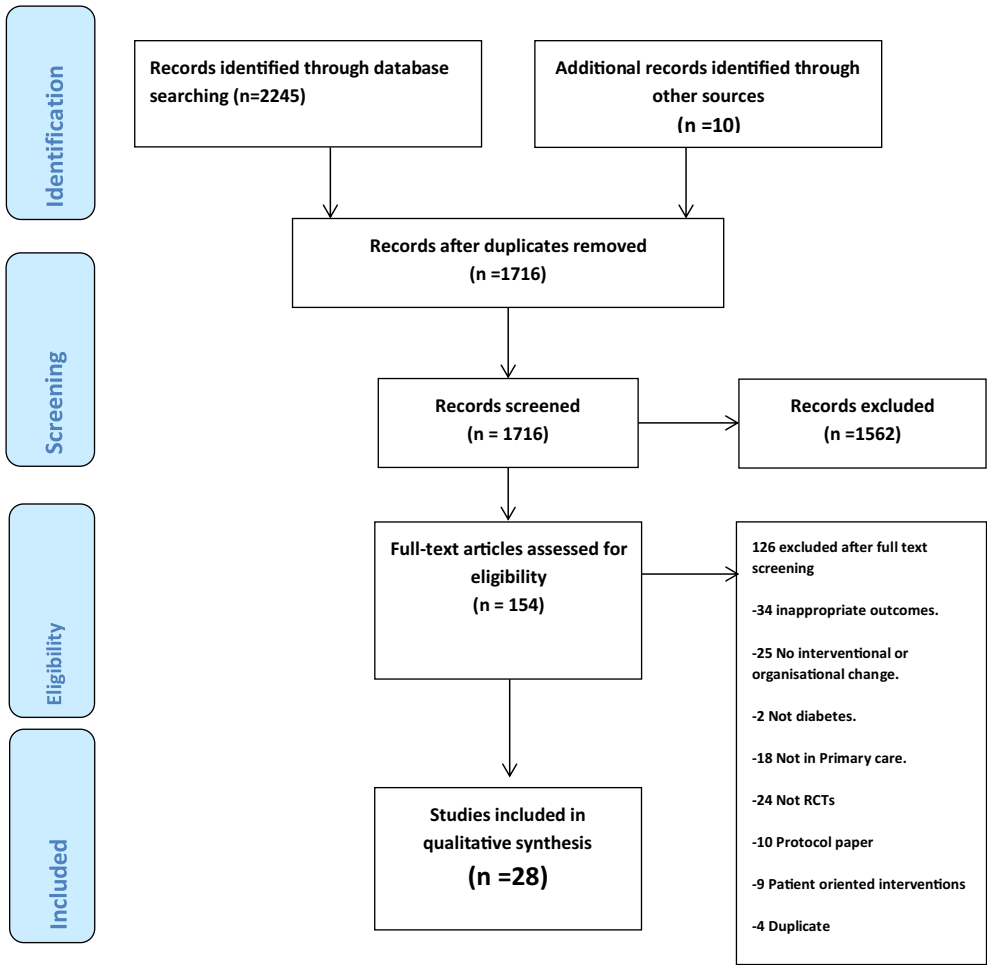

Fig. 1 – The PRISMA Flow Diagram depicting the flow of information through the different phases of a systematic review. It maps out the number of records identified, included and excluded, and the reasons for exclusions.

Table 1 – Cluster randomised trials interventions targeting general practitioners/primary health physicians.

Author	Year	Patient population	Number of patients	Interventions	Outcome	Results[a]
Piatt, [33]	2006	Underserved urban community—USA	89	Provider-based diabetes education offered to all providers via attendance at one problem based learning session	HbA1c	0
		USA			SBP	0
					LDL-C	0
Reutens, [34]	2012	Asia-Pacific region	386	Educational meetings, reminders, medical record summary sheets and patient result cards	HbA1c	0
					SBP	0
					LDL-C	0
Saenz, [35]	2012	Adult people with diabetes in Madrid—Spain	697	Computer application to help primary care professionals make decisions about the insulin handling of patients with diabetes	HbA1c	+
					LDL-C	0
Smith, [36]	2008	Mayo clinic USA	639	Specialist telemedicine intervention for improving diabetes care	HbA1C	0
					SBP	0
					LDL-C	0
Sperl-Hillen, [38]	2010	Adult patients with diabetes in Minnesota—USA	3417	Customised simulated cases with feedbacks	HbA1c	+
					SBP	0
					LDL-C	0
King, [39]	2009	People with diabetes in rural California—USA	135	Algorithm-directed care, midlevel practitioner–administered, electronically coached, treatment	HbA1c	0
					SBP	0
					LDL-C	0
Guldberg, [40]	2011	Adults with type 2 diabetes in Denmark.	2716	Electronic feedback system on type 2 diabetes population, giving them the option either to use the data during individual diabetes consultations or to gain an overview of the quality of their diabetes care and compare it with the corresponding quality in their colleagues' practices	HbA1c	0
					LDL-C	0
Foy, [41]	2011	People with diabetes, Newcastle—UK	8690	Brief educational messages, typically of less than 30 words, added to the returned results of laboratory tests ordered by clinicians on patients with diabetes	HbA1c	0
					SBP	0
					LDL-C	0

Multi-faceted interventions

Author	Year	Patient population	Number of patients	Interventions	Outcome	Results
Belary, [32]	2008	Adult patients of south Asian origin with type 2 diabetes—UK	1486	Additional time with practice nurse and support from a link worker and diabetes-specialist nurse. Prescribing support from GP and training of nurses	HbA1c	0
					SBP	0
					LDL-C	0
Olivarius, [42]	2008	Adult patients with diabetes—Denmark	848	Regular follow up and individualised goal setting supported by prompting of doctors, clinical guidelines, feedback, and continuing medical education	HbA1c	+
					SBP	+
					LDL-C	+

Table 1 (*Continued*)

Multi-faceted interventions

Author	Year	Patient population	Number of patients	Interventions	Outcome	Results
Smith, [37]	2003	Adult patients with diabetes in North Dublin—Ireland	183	Pathway redesign, 3-monthly routine reviews, Professional education, referral guidelines	HbA1c	0
					SBP	0
Peterson, [43]	2008	Adult patients with type 2 diabetes in Minnesota—USA	7101	Electronic diabetes registry, visit reminders, and patient-specific physician alerts. A site coordinator facilitated pre-visit planning and a monthly review of performance with a local physician champion	HbA1c	+
					SBP	+
					LDL-C	+
Maislos, [44]	2003	Adult patients with diabetes—Israel	82	Interdisciplinary approach	Hba1c	+
O'Connor, [45]	2005	Adult patients with type 2 diabetes. Minnesota, USA	754	Physician, a nurse, and other clinic staff interested in diabetes. Training of staff	Hba1c	0
					SBP	0
					LDL-C	0
Piatt, [33]	2006	Underserved urban Community—USA.	89	Chronic care Model	HbA1c	+
					LDL-C	+

Nurse interventions

Author	Year	Patient population	Number of patients	Interventions	Outcome	Results
Blackberry, [46]	2013	Adult patients with type 2 diabetes—Australia	473	Use of nurse for telephone coaching	HbA1c	0
					SBP	0
					LDL-C	0
Juul, [47]	2014	Adult patients with type 2 diabetes in Denmark	4034	Intervention practices were offered a 16-h self-determination theory-based course including communication training for general practice nurses delivered over 10 months	HbA1c	0
					LDL-C	0

Pharmacist interventions

Author	Year	Patient population	Number of patients	Interventions	Outcome	Results
Krais, [48]	2007	Adult patients with type 2 diabetes Australia	289	Pharmacists delivered a diabetes service to patients. They checked adherence, interactions, hypoglycaemia and lifestyle	HbA1c	+
					SBP	0
					LDL-C	0

Systolic Blood Pressure (SBP), Total Cholesterol or LDL Cholesterol (LDL-C).
[a] 0 = No statistically significant improvement, + = statistically significant effect.

cardiovascular risk factors, and addressing barriers to care. Only one study examined the role of dietician on diabetes self-management education [31]. Another considered the community health worker's role in providing culturally tailored assistance with self-management and education [30]. One study [33] had two intervention arms; a chronic care model intervention which fits our definition of a multi-faceted intervention, and healthcare professional education intervention.

Table 2 – Randomised control trials: multi-faceted intervention.

Author	Year	Population	Number of patients	Interventions	Outcome	Results
Rothman, [26]	2005	Adult patients with diabetes. USA	217	Pharmacists and care coordinator who provided diabetes education, algorithms for managing glucose and cardiovascular risk factors, and addressed barriers to care	HbA1c	+
					SBP	+
					LDL-C	0
De Pue, [22]	2013	Participants with type 2 diabetes from a community health centre in American Samoa	268	Nurse-CHW team intervention	HbA1c	+
				Staff was hired early to help conduct formative focus groups and to develop the intervention	SBP	0
Hiss, [27]		Adult patients with type 2 diabetes. Detroit USA	197	Nurse care manager with primary care physicians in patients with type 2 diabetes	HbA1c	+
					SBP	0
					LDL-C	+
Cohen, [21]	2011	Adult males with type 2 diabetes	99	Pharmacists, dieticians, nurses, and physical therapists provided educational and behavioural interventions	HbA1 C	+
					SBP	+
					LDL-C	0
Shea, [29]	2009	Adult patients with diabetes in New York-USA	1665	Telemedicine, consisting of a web-enabled computer with modem connection to an existing telephone line	HbA1c	+
					SBP	+
					LDC-L	+

Pharmacists interventions

Author	Year	Patient population	Number of patients	Interventions	Outcome	Results
Ali, [28]	2012	Adult patients with type 2 diabetes in Bedfordshire UK	46	Use of pharmacist to provide education and counselling	HbA1c	+
					SBP	+
					LDL-C	0
Clifford, [20]	2005	Adults with type 2 diabetes from the fremantle diabetes study—Australia	180	Use of pharmacist for counselling and telephone coaching	HbA1c	+
					SBP	+
Odegard, [25]	2005	Adult patients with type 2 diabetes. USA	77	Pharmacists made care plans and did weekly visits or telephone calls to facilitate diabetes management and adherence	HbA1c	0
Doucette, [23]	2009	Adults with type 2 diabetes who had completed at least 2 diabetes education sessions at a local diabetes education centre—USA	78	Discussing medications, clinical goals, and self-care activities with patients and recommending medication changes to physicians	HbA1c	0
					SBP	0

Table 2 (*Continued*)

Pharmacists interventions						
Author	Year	Patient population	Number of patients	Interventions	Outcome	Results
					LDL-C	0
Community health worker interventions						
Author	Year	Patient population	Number of patients	Interventions	Outcome	Results
Spencer, [30]	2011	African American and Latino adult participants recruited from 2 health systems in Detroit, Michigan—USA.	183	Culturally tailored CHW assistance with self-management and education	HbA1c	+
					SBP	0
					LDL-C	0
Dietician interventions						
Author	Year	Patient population	Number of patients	Interventions	Outcome	Results
Huang, [31]	2010	type 2 diabetes patients in primary care clinics in Taiwan	154	Dietician-led intervention on diabetic self-management education every 3 months over 12 months	HbA1 C	0
					SBP	+
Interventions targeting nurses						
Author	Year	Patient population	Number of patients	Interventions	Outcome	Results
Houwelling, [49]	2010	Patients with type 2 diabetes from north-east region of the Netherlands	230	Glucose levels, blood pressure and lipid profile according to a specified protocol	HbA1c	0
					SBP	0
					LDL-C	0
Interventions targeting general practitioners/primary health physicians						
Author	Year	Patient population	Number of patients	Interventions	Outcome	Results
Hermans, [50]	2011	Adult patients with type 2 diabetes in six European countries	3996	Feedback benchmarked against other centres	HbA1c	0
					SBP	+
					LDL-C	0

Systolic Blood Pressure (SBP), Total Cholesterol or LDL Cholesterol (LDL-C) 0 = No statistically significant improvement, + = statistically significant effect, Community Health Worker (CHW). Systolic Blood Pressure (SBP), Total Cholesterol or LDL Cholesterol (LDL-C) 0 = No statistically significant improvement effect, + = statistically significant effect. Systolic Blood Pressure (SBP), Total Cholesterol or LDL Cholesterol (LDL-C) 0 = No statistically significant improvement, + = statistically significant effect. Systolic Blood Pressure (SBP), Total Cholesterol or LDL Cholesterol (LDL-C) 0 = No statistically significant improvement, + = statistically significant effect. Systolic Blood Pressure (SBP), Total Cholesterol or LDL Cholesterol (LDL-C) 0 = No statistically significant improvement, + = statistically significant effect. Systolic Blood Pressure (SBP), Total Cholesterol or LDL Cholesterol (LDL-C) 0 = No statistically significant improvement, + = statistically significant effect, Community Health Worker (CHW).

Twelve studies satisfied our definition of multi-component interventions or targeted multi-disciplinary teams. [21,22,26,27,29,32,33,37,42,44,45,51]. These multi-faceted interventions ranged from multiple component interventions such as electronic coaching, staff training, algorithm driven care, reminders, alerts and audits, all in different combinations to the targeting of multi-disciplinary teams, such as nurse case managers, general practitioners, pharmacists, community health workers and dieticians.

Table 3 – Study quality properties of trials.

Study	Blinding at evaluation	Standardised reporting of outcomes	Allocation concealment described	Duration (months)
De Pue, J.D.	Not reported	Yes	Unclear	12
Doucette,W.R.	Not reported	Not reported	Unclear	12
Foy, R.	Not reported	No reported	Yes	34
Guldberg, T.L.	Not reported	Not reported	Yes	15
Hermans, M.	Not reported	No	Yes	12
Hiss, R.G.	Not reported	Yes	Unclear	6
Houwelling, S.T.	Not reported	Yes	Yes	14
Huang, M.C.	Not reported	Yes	Yes	12
Cohen, L.B.	Not reported	Not reported	Unclear	6
Clifford, R.M.	Not reported	Not reported	Unclear	12
Blackberry, I.	Yes	Not reported	Yes	18
Ali, M.	Not reported	Yes	Yes	12
King, A.B.	Not reported	Not reported	Unclear	12
Kras, I.	Not reported	Not reported	Unclear	6
O'Connor, P.J.	Not reported	No	Unclear	18
Maislos, M.	Not reported	Not reported	Unclear	6
Odegard, P.S.	Not reported	Yes	Unclear	6
Peterson, K.A.	Not reported	Yes	Unclear	12
Piatt, G.A.	Not reported	Yes	Unclear	12
Piatt, G.A.	Not reported	Yes	Unclear	12
Reutens, A.T.	Not reported	Yes	No	6
Rothman, R.L.	Not reported	Yes	Yes	12
Saenz, A.	Not reported	Not reported	Unclear	18
Shea, S.	Yes	Yes	Unclear	60
Smith, S.	Yes	Yes	Unclear	18
Smith, S.A.	Yes	Not reported	Yes	21
Spencer, M.S.	Not reported	Not reported	Unclear	6
Sperl-Hillen, J.M.	Not reported	Not reported	Unclear	12
Belary, S.	Not reported	Not reported	No	24
Olivarius, N.	Not reported	Yes	Unclear	72
Juul, L.	Not reported	Not reported	Yes	18

5. Study quality

Each of the studies had identifiable methodological limitations (Table 3). Even though all the studies included were either cluster randomised trials or randomised control trials, in 20 of them the allocation concealment was either not performed or was not reported [20–23,25,27,29,30,33–35,37–39,42,44–46,48,51]. These 20 studies did not necessarily consistently show a positive result. Some of them showed significant improvements from baseline while others did not.

Since participant blinding to intervention is not possible with the interventions under consideration, the trials were appraised based on whether researchers evaluating the outcomes were blinded to the intervention to minimise detection bias. Only four of the trials adequately described a blinded evaluation process [29,36,37,46]. Again, the presence of adequate description of blinding at the evaluation process did not necessarily predict an improvement of the outcomes from baseline as only one study out the four showed a reduction in HbA1c and SBP from baseline [29].

There was a wide range in the number of subjects in each study (Table 1); lowest number being 46 [28] and largest 8690 [41]. Smaller studies are usually likely to be underpowered to detect statistically significant effects. However, in our review there was no suggestion that studies with relatively large number of subjects necessarily produced improvements in the risk factors any more than those with small number of subjects. Out of the six studies with less than hundred subjects [21,23,28,33,44], four (66.7%) had at least one statistically significant reduction from baseline in the risk factors considered [21,28,33,44]. Out of the sixteen studies with subjects from 100 to 1000 [20,22,26,27,30,31,34–37,39,42,45,46,48,49] nine (56.3%) showed at least one statistically significant reduction in risk factor control from baseline [20,22,26,27,30,31,35,42,48]. In addition, out of the eight studies with more than one thousand subjects, [29,32,38,40,41,43,47,50], four (50%) had at least one statistically significant reduction in the outcomes from baseline [29,38,43,50].

6. Outcome effects based on the type of intervention

The detailed findings from each study were examined according to the primary care professionals targeted, the intervention methods used and the type of outcomes measured. A summary of these findings is shown in Tables 1 and 2.

Educational interventions even when they were coupled with clinician reminders did not seem to have a positive impact on the HbA1c, systolic blood pressure and lipid profiles [24,33,34,37,41].

The use of telemedicine showed mixed results based on only two studies; when used with a nurse case-manager, there were improvements in HbA1c, LDL-cholesterol and systolic blood

pressure levels compared with controls [29]. However when this intervention was directed at family practitioners with specialist advice and evidence-based messages regarding medication management for cardiovascular risk, there was no improvement in HbA1c, systolic blood pressure and LDL-cholesterol [36].

Clinician reminders in the form of a computer application to help primary care professionals make decisions about the glucose lowering therapy in diabetes patients produced significant improvements in HbA1c levels when compared to controls [35]. However algorithm-directed care and electronically coached treatment did not lead to improvements in HbA1c, systolic blood pressure or LDL-cholesterol [39]

The use of feedback to primary care professionals had a significant impact on reduction in HbA1c when it was given after a customised simulated patient interaction [38] but not when it was given after performance was benchmarked against other centres [50], even though systolic blood pressure improved. No improvement in HbA1c or cholesterol was noted compared to control when feedback was given electronically [40]

7. Effects based on primary care professional groups

Targeting general or family physicians was largely ineffective in improving the cardiovascular risk factors considered [33,34,36,39–41,50], except when using a computer application on insulin handling of type 2 diabetes or customised simulated cases with feedbacks [35,38]. Similarly, interventions targeting nurses did not improve outcomes compared to standard care [24,46,49].

The use of pharmacists produced mixed results in terms of improving the outcomes in the intervention groups. HbA1c improvement was not achieved in the intervention group by pharmacists discussing medications, clinical goals, and self-care activities with patients and recommending medication changes to physicians when appropriate [23]. The use of clinical pharmacists to establish and initiate a diabetes care plan followed by weekly visits or telephone calls to facilitate diabetes management and adherence also did not lead to improvements in HbA1c control [25]. However, the use of clinical pharmacists for counselling, patient education and telephone coaching as well as management and regular reviews to support self-monitoring of blood glucose, adherence support, and reminders of checks for diabetes complications led to improvements in HbA1c [20,28,48]. The use of a clinical pharmacist to provide education and counselling to patients also showed mixed results; improvements in systolic blood pressure was noted when compared to controls in two studies [20,28] although two other studies showed no improvements in blood pressure [23,48]. In terms of improvement in cholesterol from baseline, when compared to controls, pharmacist interventions have largely not been successful [23,28,48].

The use of a registered dietician to provide on-site diabetic self-management education every 3 months over 12 months failed to lead to improvements in HbA1c [31].

Using community health workers to provide a culturally tailored assistance with self-management and education led to reductions in HbA1C but not systolic blood pressure or lipids [22,30].

8. Multi-component interventions and multi-disciplinary organisational changes

Eight of the twelve studies that considered multi-disciplinary organisational changes mixed with various interventions reported improvements in HbA1c [21,22,26,27,33,42–44]. These included clinically significant reductions of 1.8% [52], 0.8% [26], 0.6% [33], and 0.53% [22] from baseline. The improvements in systolic blood pressure from baseline when compared to control groups were however inconclusive; four studies reporting improvements in systolic blood pressure [21,26,42,51] and another three failing to show any improvements [22,27,33]. Similarly, improvements in lipids were also inconclusive; five studies [21,26,32,42,45] showing no effect and another three showing an improvement [27,33,43].

9. Discussion

Multi-component interventions generally appear to lead to statistically and clinically significant improvements in HbA1c in people with diabetes as opposed to mono-component interventions. This finding was typified by the one study that compared a multi-faceted intervention (Chronic Care Model) and professional education to standard care [33]. Both the professional education and standard care interventions did not show improvements in HbA1c; however the multifaceted interventions showed a marked decline in HbA1c. Practitioner education even when combined with reminders did not demonstrate a significant improvement in the levels of HbA1c, blood pressure and LDL-cholesterol from baseline. This finding is consistent with earlier publications on this subject [53]. The lack of efficacy of general practitioner education has been attributed to the fact that the generalists tend to consider factors such as the diabetes management costs, insufficient time, lack of financial incentives and patient refusal of insulin as barriers to effective control of diabetes [34].

Similarly the use of other mono-component interventions like feedback, clinical reminders, and telemedicine alone have not consistently led to improvements in the HbA1c, SBP and total or LDL-cholesterol possibly because they are delivered in isolation without much organisational support to enhance patient-provider interactions. Studies in which these mono-component interventions have yielded some improvements in cardiovascular risk factors had other components of the multi-faceted interventions like the use of case managers trained in diabetes management [29]. Even the Quality and Outcomes Framework in the UK which is one of the world's largest pay-for-performance schemes, rewarding general practitioners for the quality of care they provide, only had limited impact on improving health outcomes [54,55]. The coupling of customised patient specific feedback bench-marked against the performance of other clinicians and financial incentives led to improvements in outcomes [38]. The organisation of care as a whole delivery system design with the revision of roles of primary care professionals, aided with decision support, and information technology and other community linkages should be done in a structured manner in order to improve outcomes. Improving the cooperation between various primary care

professionals is likely to be effective if each team member works to achieve improvements in similar outcomes. Previous studies have reported that interactive education, audit and feedback, reminders, academic detailing and other outreach visits were the most effective in changing physician care and patient outcomes [10,56,57], whereas simple distribution of clinical practice guidelines and use of opinion leaders were less effective [56].

Since we have not found any consistency in association of improved cardiovascular risks and the types of primary care professionals used at the forefront of diabetes care, the use of practice nurses, pharmacists, dieticians or general practitioners could all be considered in various settings provided the organisation and delivery of diabetes care is done in structured manner. In areas where there are more nurses than other primary care professionals it will be better to use them as case managers aided by telemedicine [29]. When community pharmacists have to be used, it is better to combine their educational updates with regular patient reviews counselling of patients on self-monitoring of blood glucose, adherence support, and reminders of checks for diabetes complications [20,28,48]. Interventions targeting patients have not been considered in this review as there is ample evidence on the effectiveness of these interventions on various health outcomes. It is therefore reasonable to consider advocating patient targeted interventions in combination with the multi-faceted health care professional interventions in order to maximise improvements in outcomes.

10. Strengths and limitations

To our knowledge this is the first attempt to consider both primary care and community based interventions in a systematic review on diabetes patients. We identified a large number of studies from multiple databases. We appraised the quality of the studies included in a systematic manner.

Several factors may affect the study outcomes thus limiting applicability of our findings in clinical practice. First, the rigor and quality of the various interventions are questionable. Like most complex interventions, the chances of practitioners consulting with colleagues and networking and thus increasing the risk of contamination in the standard care groups can occur.

Second, we attempted as much as we could to get a homogenous number of studies for this review with clear inclusion and exclusion criteria in terms of the study types and outcomes. However there were still wide variations in the population of patients and target health care professions. Even the interventions studied varied widely in terms of duration, delivery methods and care systems. This could affect the generalisability of the review.

Third our review is limited by the quality of the studies included. Inadequate information about concealment of allocation, lack of standardised reporting of the outcomes, inadequate information on blinding at outcome evaluation and wide differences in length of follow ups are markers of poor qualities in the studies.

Also, it is possible that if the patient targeted interventions were considered in this review, the generally negative results seen in the mono-component health care professional interventions would have been positive. This is because the lack of statistically significant improvements in the risk factors could be as a result of lack of engagement from patients.

Lastly, our literature search did not identify any unpublished studies, thus raising the possibility of publication bias in some of the studies that reported improvements in outcomes.

11. Implication for practice

There was a lack of good quality evidence of cardiovascular risk factor improvement when mono-component interventions were targeted at primary care or community based professionals, irrespective of their roles. When planning diabetes implementation programmes in primary care settings, it is therefore important to avoid using single standalone methods such as healthcare professional education, telemedicine, clinician reminders and audits. This approach can be expensive to deliver and may not yield the desired outcomes. Instead a collaborative approach, involving the use of multifaceted interventions targeting various professionals in a structured fashion should be considered in all cases. This systematic review shows that a complex structure will yield improvements in outcomes when customised to the needs of local populations. For example, in the primary care setting the use of pharmacists in areas where this resource abounds can yield improvements in diabetes intermediate outcomes provided this is done in a structured collaborative manner.

12. Conclusion

This systematic review on interventions targeting primary care or community based professionals on diabetes and other cardiovascular risk factor control has not shown consistency for single stranded interventions on the healthcare professionals. Multi-disciplinary team collaborations mixed with various interventions reported some consistency in improvements in HbA1c.

Funding

SS, NW, DB, MJD and KK acknowledge support from the National Institute for Health Research Collaboration for Leadership in Applied Health Research and Care–East Midlands (NIHR CLAHRC–EM), the Leicester Clinical Trials Unit and the NIHR Leicester-Loughborough Diet, Lifestyle and Physical Activity Biomedical Research Unit, which is a partnership between University Hospitals of Leicester NHS Trust, Loughborough University and the University of Leicester.

Declarations

SS has received honoraria for speaking at meetings and serving on Advisory Boards for Novartis, Novo Nordisk, Janssen, MSD, Astra Zeneca, Sanofi Aventis, Lilly and BI.

MJD has acted as consultant, advisory board member and speaker for Novo Nordisk, Sanofi-Aventis, Lilly, Merck Sharp &

Dohme, Boehringer Ingelheim, AstraZeneca and Janssen and as a speaker for Mitsubishi Tanabe Pharma Corporation. She has received grants in support of investigator and investigator initiated trials from Novo Nordisk, Sanofi-Aventis and Lilly.

KK has acted as consultant, advisory board member and speaker for Astra Zeneca, Lilly, Novartis, Boehringer Ingelheim, Janssen, Sanofi Aventis, MSD and Novo Nordisk.

Conflict of interest statement

NSW and DHB have no conflict of interest.

Contributorship statement

The analysis was planned and managed by KK and SS. NSW acted as a second reviewer, DHB assisted with protocol design and data extraction. DHB, NSW, KK and MJD, commented on drafts of the paper. All authors gave final approval of the version to be published. The views expressed in this article are the authors', and are independent of the NIHR.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.diabres.2016.01.022.

REFERENCES

[1] Guariguata L, Whiting D, Hambleton I, Beagley J, Linnenkamp U, Shaw J. Global estimates of diabetes prevalence for 2013 and projections for 2035. Diabetes Res Clin Pract 2014;103(2):137–49.

[2] Whiting DR, Guariguata L, Weil C, Shaw J. IDF diabetes atlas: global estimates of the prevalence of diabetes for 2011 and 2030. Diabetes Res Clin Pract 2011;94(3):311–21.

[3] Khunti K, Ganguli S. Who looks after people with diabetes: primary or secondary care? J R Soc Med 2000;93(4):183–6.

[4] Martin Roland D. Linking physicians' pay to the quality of care—a major experiment in the United Kingdom. N Engl J Med 2004;351:1448–54.

[5] Doran T, Kontopantelis E, Valderas JM, Campbell S, Roland M, Salisbury C, et al. Effect of financial incentives on incentivised and non-incentivised clinical activities: longitudinal analysis of data from the UK quality and outcomes framework. BMJ 2011;342(Jun):d3590.

[6] Millett C, Netuveli G, Saxena S, Majeed A. Impact of pay for performance on ethnic disparities in intermediate outcomes for diabetes: a longitudinal study. Diabetes Care 2009;32(3):404–9.

[7] Millett C, Bottle A, Ng A, Curcin V, Molokhia M, Saxena S, et al. Pay for perfomance and the quality of diabetes management in individuals with and without co-morbid medical conditions. J R Soc Med 2009;102(9):369–77.

[8] Kogan AJ. Overcoming obstacles to effective care of type 2 diabetes. Am J Manag Care 2009;15(9 Suppl):S255–62.

[9] Stone MA, Charpentier G, Doggen K, Kuss O, Lindblad U, Kellner C, et al. Quality of care of people with type 2 diabetes in eight European countries: findings from the guideline adherence to enhance care (GUIDANCE) study. Diabetes Care 2013;36(9):2628–38.

[10] Davis D. Does CME work? An analysis of the effect of educational activities on physician performance or health care outcomes. Int J Psychiatry Med 1998;28(1):21–39.

[11] Davis D, O'Brien MAT, Freemantle N, Wolf FM, Mazmanian P, Taylor-Vaisey A. Impact of formal continuing medical education: do conferences, workshops, rounds, and other traditional continuing education activities change physician behavior or health care outcomes? JAMA 1999;282(9):867–74.

[12] Davis DA, Barnes BE, Fox RD. The continuing professional evelopment of physicians: from research to practice. Chicago: American Medical Association Press; 2003.

[13] Cook DA, Levinson AJ, Garside S, Dupras DM, Erwin PJ, Montori VM. Internet-based learning in the health professions: a meta-analysis. JAMA 2008;300(10):1181–96.

[14] Renders CM, Valk GD, Griffin SJ, Wagner EH, Eijk Van JT, Assendelft WJ. Interventions to improve the management of diabetes in primary care, outpatient, and community settings: a systematic review. Diabetes Care 2001;24(10):1821–33.

[15] Davies MJ, Heller S, Skinner TC, Campbell MJ, Carey ME, Cradock S, et al. Effectiveness of the diabetes education and self management for ongoing and newly diagnosed (DESMOND) programme for people with newly diagnosed type 2 diabetes: cluster randomised controlled trial. BMJ 2008;336(Mar (7642)):491–5.

[16] Khunti K, Gray LJ, Skinner T, Carey ME, Realf K, Dallosso H, et al. Effectiveness of a diabetes education and self management programme (DESMOND) for people with newly diagnosed type 2 diabetes mellitus: three year follow-up of a cluster randomised controlled trial in primary care. BMJ 2012;344(Apr):e2333.

[17] Cooper H, Booth K, Gill G. A trial of empowerment-based education in type 2 diabetes—global rather than glycaemic benefits. Diabetes Res Clin Pract 2008;82(2):165–71.

[18] Deakin T, Cade J, Williams R, Greenwood D. Structured patient education: the Diabetes X-PERT programme makes a difference. Diabetic Med 2006;23(9):944–54.

[19] Howorka K, Pumprla J, Wagner-Nosiska D, Grillmayr H, Schlusche C, Schabmann A. Empowering diabetes out-patients with structured education: short-term and long-term effects of functional insulin treatment on perceived control over diabetes. J Psychosom Res 2000;48(1):37–44.

[20] Clifford RM, Davis WA, Batty KT, Davis TM. Effect of a pharmaceutical care program on vascular risk factors in type 2 diabetes: the Fremantle Diabetes Study. Diabetes Care 2005;28(4):771–6.

[21] Cohen LB, Taveira TH, Khatana SA, Dooley AG, Pirraglia PA, Wu WC. Pharmacist-led shared medical appointments for multiple cardiovascular risk reduction in patients with type 2 diabetes. Diabetes Educ 2011;37(6):801–12.

[22] De Pue JD, Dunsiger S, Seiden AD, Blume J, Rosen RK, Goldstein MG, et al. Nurse-community health worker team improves diabetes care in American Samoa: results of a randomized controlled trial. Diabetes Care 2013;36(7):1947–53.

[23] Doucette WR, Witry MJ, Farris KB, McDonough RP. Community pharmacist-provided extended diabetes care. Ann Pharmacother 2009;43(5):882–9.

[24] Juul L, Maindal HT, Zoffmann V, Frydenberg M, Sandbaek A. Effectiveness of a training course for general practice nurses in motivation support in type 2 diabetes care: a cluster-randomised trial. PLoS ONE 2014;9(5):e96683 [Electronic Resource].

[25] Odegard PS, Goo A, Hummel J, Williams KL, Gray SL. Caring for poorly controlled diabetes mellitus: a randomized pharmacist intervention. Ann Pharmacother 2005;39(3):433–40.
[26] Rothman RL, Malone R, Bryant B, Shintani AK, Crigler B, Dewalt DA, et al. A randomized trial of a primary care-based disease management program to improve cardiovascular risk factors and glycated hemoglobin levels in patients with diabetes. Am J Med 2005;118(3): 276–84.
[27] Hiss RG, Armbruster BA, Gillard ML, McClure LA. Nurse care manager collaboration with community-based physicians providing diabetes care a randomized controlled trial. Diabetes Educ 2007;33(May/June (3)):493–502.
[28] Ali M, Schifano F, Robinson P, Phillips G, Doherty L, Melnick P, et al. Impact of community pharmacy diabetes monitoring and education programme on diabetes management: a randomized controlled study. Diabetic Med 2012;29(9):e326–33.
[29] Shea S, Weinstock RS, Teresi JA, Palmas W, Starren J, Cimino JJ, et al. A randomized trial comparing telemedicine case management with usual care in older, ethnically diverse, medically underserved patients with diabetes mellitus: 5 year results of the IDEATel study. J Am Med Inf Assoc: JAMIA 2009;16(4):446–56.
[30] Spencer MS, Rosland AM, Kieffer EC, Sinco BR, Valerio M, Palmisano G, et al. Effectiveness of a community health worker intervention among African American and Latino adults with type 2 diabetes: a randomized controlled trial. Am J Public Health 2011;101(12):2253–60.
[31] Huang MC, Hsu CC, Wang HS, Shin SJ. Prospective randomized controlled trial to evaluate effectiveness of registered dietitian-led diabetes management on glycemic and diet control in a primary care setting in Taiwan. Diabetes Care 2010;33(2):233–9.
[32] Bellary S, O'Hare J, Raymond N, Gumber A, Mughal S, Szczepura A, et al. Enhanced diabetes care to patients of south Asian ethnic origin (the United Kingdom Asian Diabetes Study): a cluster randomised controlled trial. Lancet 2008;371(9626):1769–76. 20080524.
[33] Piatt GA, Orchard TJ, Emerson S, Simmons D, Songer TJ, Brooks MM, et al. Translating the chronic care model into the community: results from a randomized controlled trial of a multifaceted diabetes care intervention. Diabetes Care 2006;29(4):811–7.
[34] Reutens AT, Hutchinson R, Binh T, Cockram C, Deerochanawong C, Ho LT, et al. The GIANT study, a cluster-randomised controlled trial of efficacy of education of doctors about type 2 diabetes mellitus management guidelines in primary care practice. Diabetes Res Clin Pract 2012;98(1):38–45.
[35] Saenz A, Brito M, Moron I, Torralba A, Garcia-Sanz E, Redondo J. Development and validation of a computer application to aid the physician's decision-making process at the start of and during treatment with insulin in type 2 diabetes: a randomized and controlled trial. J Diabetes Sci Technol 2012;6(May (3)):581–8.
[36] Smith SA, Shah ND, Bryant SC, Christianson TJ, Bjornsen SS, Giesler PD, et al. Chronic care model and shared care in diabetes: randomized trial of an electronic decision support system. Mayo Clin Proc 2008;83(7):747–57.
[37] Smith SM, Bury G, O'Leary M, Shannon W, Tynan A, Staines A, et al. The North Dublin randomized controlled trial of structural diabetes shared care. Fam Pract 2004;21(1):39–45.
[38] Sperl-Hillen JM, O'Connor PJ, Rush WA, Johnson PE, Gilmer T, Biltz G, et al. Simulated physician learning program improves glucose control in adults with diabetes. Diabetes Care 2010;33(8):1727–33.
[39] King AB, Wolfe GS. Evaluation of a diabetes specialist-guided primary care diabetes treatment program. J Am Acad Nurse Pract 2009;21(1):24–30.
[40] Guldberg TL, Vedsted P, Kristensen JK, Lauritzen T. Improved quality of Type 2 diabetes care following electronic feedback of treatment status to general practitioners: a cluster randomized controlled trial. Diabetic Med 2011;28(3):325–32.
[41] Foy R, Eccles MP, Hrisos S, Hawthorne G, Steen N, Gibb I, et al. A cluster randomised trial of educational messages to improve the primary care of diabetes. Implement Sci 2011;6:129.
[42] Olivarius NF, Beck-Nielsen H, Andreasen AH, Horder M, Pedersen PA. Randomised controlled trial of structured personal care of type 2 diabetes mellitus. BMJ 2001;323(Oct (7319)):970–5.
[43] Peterson KA, Radosevich DM, O'Connor PJ, Nyman JA, Prineas RJ, Smith SA, et al. Improving diabetes care in practice: findings from the TRANSLATE trial. Diabetes Care 2008;31(12):2238–43.
[44] Maislos M, Weisman D. Multidisciplinary approach to patients with poorly controlled type 2 diabetes mellitus: a prospective, randomized study. Acta Diabetol 2004;41(June (2)):44–8.
[45] O'Connor PJ, Desai J, Solberg LI, Reger LA, Crain AL, Asche SE, et al. Randomized trial of quality improvement intervention to improve diabetes care in primary care settings. Diabetes Care 2005;28(8):1890–7.
[46] Blackberry ID, Furler JS, Best JD, Chondros P, Vale M, Walker C, et al. Effectiveness of general practice based, practice nurse led telephone coaching on glycaemic control of type 2 diabetes: the patient engagement and coaching for health (PEACH) pragmatic cluster randomised controlled trial. BMJ 2013;347(7926):f5272 (Online).
[47] Juul L, Maindal HT, Zoffmann V, Frydenberg M, Sandbaek A. A cluster randomised pragmatic trial applying self-determination theory to type 2 diabetes care in general practice. BMC Fam Pract 2011;12:130.
[48] Krass I, Armour CL, Mitchell B, Brillant M, Dienaar R, Hughes J, et al. The pharmacy diabetes care program: assessment of a community pharmacy diabetes service model in Australia. Diabetic Med 2007;24(6): 677–83.
[49] Houweling ST, Kleefstra N, Hateren KJ, Groenier KH, Meyboom-de Jong B, Bilo HJ. Can diabetes management be safely transferred to practice nurses in a primary care setting? A randomised controlled trial. J Clin Nurs 2011;20(9–10):1264–72.
[50] Hermans M, Muls E, Nobels F, Krzentowski G, Claes N, Debacker N, et al. Evaluating benchmarking to optimize management of type 2 diabetes patients: the Belgian data from the optimise study. Acta Cardiol 2011;66(1): 124–5.
[51] Peterson KA, Radosevich DM, O'Connor P, Nyman JA, Prineas RJ, Smith SA, et al. Improving diabetes care in practice. Diabetes Care 2008;31(12):2238–43.
[52] Maislos M, Weisman D. Multidisciplinary approach to patients with poorly controlled type 2 diabetes mellitus: a prospective, randomized study. Acta Diabetol 2004;41(Jun (2)):44–8.
[53] Thepwongsa I, Kirby C, Schattner P, Shaw J, Piterman L. Type 2 diabetes continuing medical education for general practitioners: what works? A systematic review. Diabetic Med 2014;31(12):1488–97.
[54] Langdown C, Peckham S. The use of financial incentives to help improve health outcomes: is the quality and outcomes framework fit for purpose? A systematic review. J Public Health (Oxf) 2014;36(Jun (2)):251–8.

[55] Khunti K, Gadsby R, Millett C, Majeed A, Davies M. Quality of diabetes care in the UK: comparison of published quality of care reports with results of the quality and outcomes framework for diabetes. Diabetic Med 2007;24(12):1436–41.
[56] Oxman AD, Thomson MA, Davis DA, Haynes RB. No magic bullets: a systematic review of 102 trials of interventions to improve professional practice. CMAJ 1995;153(Nov (10)): 1423–31.
[57] Sohn W, Ismail AI, Tellez M. Efficacy of educational interventions targeting primary care providers' practice behaviors: an overview of published systematic reviews. J Public Health Dent 2004;64(3):164–72.